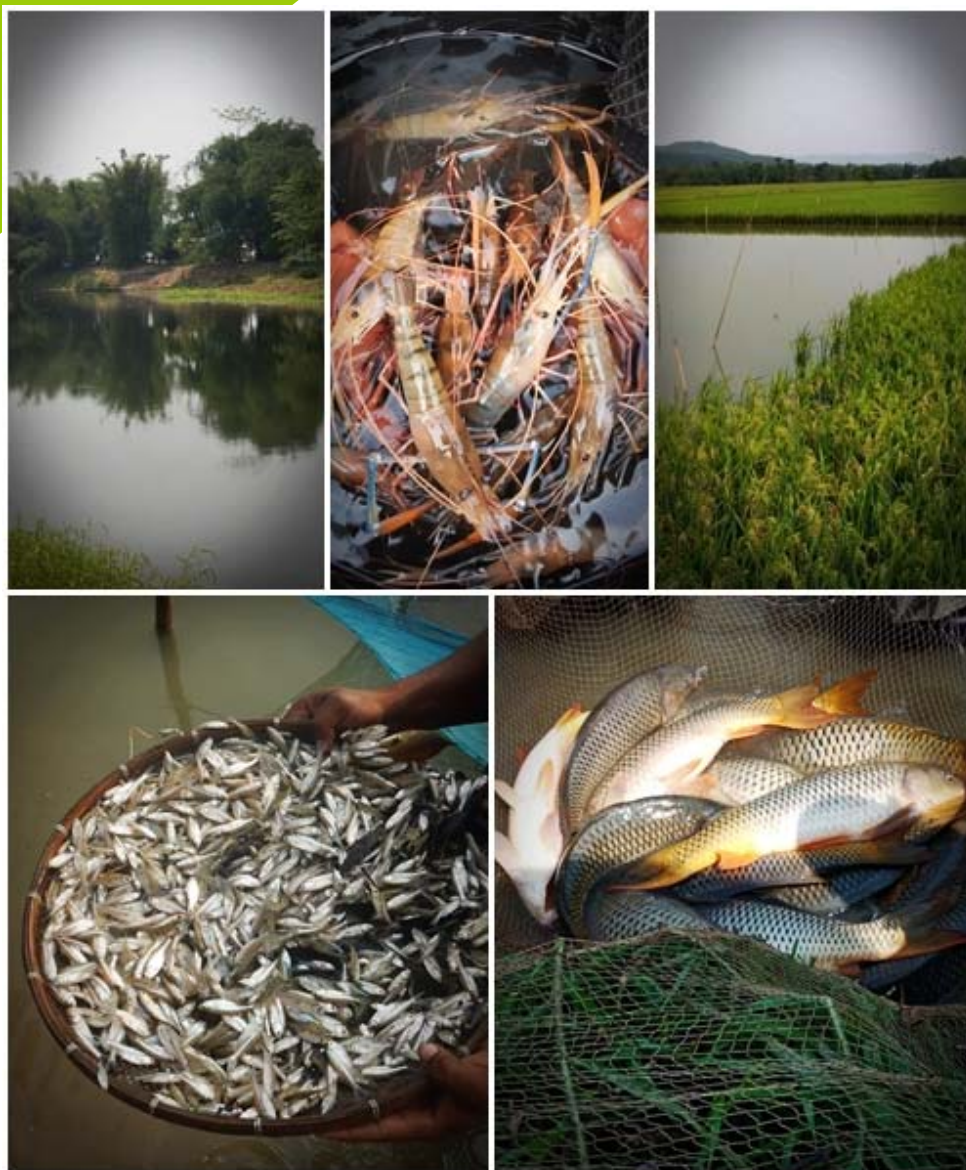




Trainers Manual

**Better Management Practices in Fisheries
Interventions under APART**

Complied by

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Funded by





Trainers Manual

**Better Management Practices in
Fisheries Interventions**

under

**Assam Agribusiness and Rural
Transformation Project**

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Lists of Abbreviations

APART	Assam Agribusiness and Rural Transformation Project
ARIASS	Assam Rural Infrastructure and Agricultural Services Society
PDO	Project development objective
BMP	Better Management Practices
ToT	Training of Trainers
KVK	Krishi Vigyan Kendra
N	Nitrogen
P	Phosphorus
Kg	Kilogram
g	Gram
ha	Hectare
m	Meter
cm	Centimeter
KMnO ₄	Potassium permanganate
NaCl	Sodium chloride
DO	Dissolved Oxygen
Nos	Numbers
H ₂ S	Hydrogen Sulfide
(SIS)	Small indigenous fish species
CP	Crude protein
ATMA	Agricultural Technology Management Agency
AAU	Assam Agricultural University
IRRI	International Rice Research Institute
SOP	Standard Operating Procedure
ml	Milli liter
PPM	Parts per million
P ₂ O ₅	Phosphorus pentoxide
K ₂ O	Potassium oxide
FYM	Farmyard manure

Preface

The Government of Assam, through the Government of India, has received a loan of USD 200 million from the World Bank for implementation of the Assam Agribusiness and Rural Transformation Project (APART). The project development objective (PDO) is to add value and improve resilience of selected agriculture value chains, focusing on smallholder farmers and Agro entrepreneurs in targeted districts of Assam. Fisheries is one the prioritized value chains for interventions under APART for sustainable increase in fish production and productivity through climate resilient smart aquaculture technologies in market led production approach. The fisheries value chain is implemented by the Department of Fisheries, Assam in close coordination with Assam Rural Infrastructure and Agricultural Services (ARIAS) Society, private sector partners with technical support from WorldFish, Malaysia.

WorldFish is providing technical support to the ARIAS Society, Department of Fisheries and College of Fisheries, Assam Agricultural University in the implementation of the Fisheries Sub-Component of the APART project in Assam from Aug 2018. Keeping in line with the Project Development Objective of APART, the WorldFish technical support envisages accomplishing the following five broad objectives:

1. Enable sustainable increases in aquaculture production without creating adverse socio economic or environmental impacts (Sustainable Intensification of Aquaculture)
2. Secure and enhance the contribution of small-scale fisheries to food security in Assam (Increasing Diversity and Productivity of Beels)
3. Increase the availability, access and consumption of nutrient-rich, safe fish, especially for women of reproductive age, infants and young children (Improving Fish Value Chains and human nutrition)
4. Develop and promote climate resilient technologies in support of sustainable aquaculture and small-scale fisheries (Climate resilient/smart aquaculture technologies)
5. Promote gender transformative approaches in support of sustainable aquaculture and beel fisheries in Assam (Gender Transformative Approaches in aquaculture and fisheries)

Developing and promoting adoption of Better Management Practices for sustainable intensification of aquaculture is one of the key deliverables by WorldFish under the

APART project. Better Management Practices refers to a set of management practices that are developed based on the existing practices and associated risks as determined in consultation with scientists and farmers and relevant industry stakeholders. BMP development process considers the technical, environmental, social and economic issues associated with farming practices and are always location and context specific. BMPs are dynamic and evolving over period and need to be updated periodically. BMPs would help to improve productivity, reduce negative environmental and social consequences of farming. WorldFish has developed the first version of different fisheries interventions under APART in consultation with various stakeholders of fisheries and aquaculture sector in Assam which include:

1. Better Management Practices for Carp Polyculture.
2. Better Management Practices for Production Mola and SIS along with Carps.
3. Better Management Practices for Polyculture of Carps with Genetically Improved Fish Strains.
4. Better Management Practices for Polyculture of Carps with Freshwater Prawns.
5. Better Management Practices for Paddy cum Fish Integrated Farming.
6. Better Management Practices for Indian Major Carp Hatcheries for Quality Seed Production.
7. Better Management Practices for Beel Fisheries.

The BMP practices for the various interventions in the fisheries sub-component of APART have been printed in local language and distributed for adoption by the farmers. For effective dissemination of the BMP practices for large scale adoption by the farmers, there is need to provide training on the BMP practices for the beneficiaries and non-beneficiary farmers in various APART districts. College of Fisheries, Assam Agricultural University, Raha has taken up the herculean task of providing various capacity building programs of various interventions in the fisheries sub-component of Assam. To implement a successful training programme, training the trainers with necessary knowledge and skill is crucial. Training of Trainers (ToT) programs were organised by the College of Fisheries to train the extension personnel from Department of Fisheries, KVKs and APART Project Team. In addition to the ToT programs, WorldFish has prepared a Trainers Manual of the various BMP Practices to help the trainers to conduct the training programs under APART. To maintain the quality of training and uniformity of technical content, this trainer's manual will be useful for the trainers. This trainer's manual is complementary to Better Management Practices developed by WorldFish, to provide a systematic guidance to the trainers on the technical content to be delivered while conducting training programs.

Chapter 1



Polyculture of carps



1 Polyculture of carps

Introduction

Fish Polyculture is the system in which fast growing compatible species of different feeding habits and possibly of different age groups of the same species, are stocked together in the same water body (ponds, lakes etc) so that all its ecological niches are occupied by fish. Main target is to obtain high production from unit of waterbody. Fast growing compatible species of different feeding habit are stocked in different proportion in the same pond so that all its ecological niches are occupied. The maximum utilization of all the requirements of life takes place without doing any harm to each other. Polyculture of carps is widely practiced by the fish farmers in Assam.

Pond Preparation

- ❖ Drain pond; remove sludge from the bottom and let it dry
- ❖ Ploughing of pond bottom
- ❖ Keep pond dyke clean from bushes to prevent hiding of predators
- ❖ Repair dike as necessary to control water seepage
- ❖ Repair inlet and outlet pipes
- ❖ Place bamboo or other substrates – which help in colonizing periphyton and help in protecting from poaching of fish
- ❖ Apply lime, distributing evenly over the pond surface, at a recommended rate
- ❖ Consider fencing the sides of pond using nylon net/bamboo/any other locally available materials or bio-fencing using plants like Assam Lemon, Citronella

Water Filling and fertilization of pond

- ❖ Fill pond; maintain water depth between 1.5-2.0 m. Filter water with double lined nylon nets if water from natural sources is used for filling.
- ❖ Fertilize pond, using chemical fertilizers @28 kg N and 21 kg P/ha with Urea and single super phosphate OR other fertilizers readily available in local market.
- ❖ Applying recommended doses of organic manure (Raw Cow Dung @10000 kg/ha) in heaps in one or two pond corners enclosed with the help of split bamboo.

Stocking

- ❖ Pond is ready to stock once water turns green (rich in plankton – natural food for fish)

- ❖ Purchase fish seeds from Department hatcheries, Multiplication centres or Registered Private hatcheries. Stocking of genetically improved Jayanti rohu and Amur carp is highly recommended
- ❖ Treat the fish seeds with KMnO_4 @1-3 ppm or NaCl @4% before stocking
- ❖ Maintain stocking density (8000 nos/ha) as recommended
- ❖ Maintain stocking ratio of 4:3:3 (surface feeder: column feeder: bottom feeder)

Water quality management

- ❖ Monitor the pond regularly (early morning and late in the afternoon):
- ❖ Low DO stress is likely:
 - if the pond is dark green with Secchi disc visibility of <15 cm (DO stress occurs either very early in the morning or when the day is cloudy)
 - if pond water is dark/black coloured with foul smell (eg., H_2S)
- ❖ Check pH after heavy rainfall and apply lime if pH is <6.5

Feeding

- ❖ Follow feeding options, depending on the feed availability and costs:
 - Floating pellets
 - Sinking pellets
 - Farm-made feed
 - Combination of commercial formulated feed and farm-made feed
- ❖ Adjust feeding rate as recommended

Regular Sampling

- ❖ Do sampling every month to assess the growth and health of the cultured fishes:
 - Monitor the length and weight of different species of fishes stocked.
 - Use the information for adjusting the feeding rates.
 - Check for the presence of external parasites on the body surface, fins, gills etc. and for the presence of lesions or wounds.

Disease Management

- ❖ There are a numbers of preventive measures farmers can use to reduce the risk of diseases and parasitic infection:
 - Stock pond with healthy, disease-free seed.
 - Disinfect all tools used in fish culture operations.
 - Avoid overcrowding.
 - Avoid restocking the ponds with fishes from unknown sources. If unavoidable, check the health of fishes stocked
 - Follow proper feeding, fertilization and water quality management practices.

- Conduct routine screening for any sign of disease/stress.
- Record all clinical signs; take pictures of clinical signs, and record of number of dead fish.
- Remove all moribund and dead fish and bury them far away from the pond.
- Bring any unusual mortality to the attention of responsible authorities and assist them to collect sick and moribund samples for laboratory testing.
- Follow the suggestion of the responsible authority for disease management

Harvesting

- ❖ Follow harvesting strategy based on culture system – generally single stocking and multiple harvesting is recommended:
 - Harvest fishes when they grow to 800g -1 kg. size
 - Condition fishes for 4 to 6 hours before sending to live fish market

Record Keeping

- ❖ Maintain records of all farm activities, stocking details, inputs applied, sampling details, harvest details, sales and consumption.
- ❖ Use record books provided under the project for keeping records.
- ❖ Ensure the records are updated periodically by the Project staff or Department officials during their visit to the farm.

Chapter 2



Polyculture of carps with mola and other Small Indigenous Fish Species (SIS)



2 Polyculture of carps with mola and other Small Indigenous Fish Species (SIS)

Introduction

Fish Polyculture is the system in which fast growing compatible species of different feeding habits and possibly of different age groups of the same species, are stocked together in the same water body (ponds, lakes etc) so that all its ecological niches are occupied by fish. Main target is to obtain high production from unit of waterbody. Fast growing compatible species of different feeding habit are stocked in different proportion in the same pond so that all its ecological niches are occupied. The maximum utilization of all the requirements of life takes place without doing any harm to each other. Polyculture of carps is widely practiced by the fish farmers in Assam.

Assam has abundant resources of small indigenous fish species (SIS) rich in micronutrients such as vitamin A, calcium, Iron, zinc etc. Among the several species of SIS, *Amblypharyngodon mola* locally called *Moa* is a nutrient dense fish, it contains vitamin-A, Iron, Zinc, and Calcium and these nutrients are highly required for pregnant and lactating women as well as young children. As mola is a potential candidate species and can be cultured along with carps in pond polyculture without hampering the target species, carp-mola-SIS polyculture technology is demonstrated under the project. The mola and SIS production technology can provide an additional production and income to the farmers and regular consumption of mola and SIS help to eradicate malnutrition and provide nutritional security for the local communities.

Pond Preparation

- ❖ Drain pond; remove sludge from the bottom and let it dry
- ❖ Ploughing of pond bottom
- ❖ Keep pond dyke clean from bushes to prevent hiding of predators
- ❖ Repair dyke as necessary to control water seepage
- ❖ Repair inlet and outlet pipes
- ❖ Place bamboo or other substrates – which help in colonizing periphyton and also help in protecting from poaching of fish
- ❖ Apply lime, distributing evenly over the pond surface, at a recommended rate
- ❖ Consider fencing the sides of pond using nylon net/bamboo/any other locally available materials or bio-fencing using plants like Assam Lemon, Chitranalla

Water Filling and fertilization of pond

- ❖ Fill pond; maintain water depth between 1.5-2.0 m. Filter water with double lined nylon nets if water from natural sources is used for filling.
- ❖ Fertilize pond, using chemical fertilizers @28 kg N and 21 kg P/ha with Urea and single super phosphate OR other fertilizers readily available in local market.
- ❖ Applying recommended doses of organic manure (Raw Cow Dung @10000 kg/ha) in heaps in one or two pond corners enclosed with the help of split bamboo.

Identification of Mola and SIS broodfish source

- ❖ Mola and SIS are self- recruiting fish species inhabiting rivers, streams, beels, ponds, ditches, tanks and inundated paddy fields. It is commonly found in larger ponds connected to open rice fields and ditches in the floodplains.
- ❖ The mola and other SIS brood source need to be initially identified. Ideally, fish farming clusters should have good records of mola and SIS source that can be used for collection of matured broodfish. It is also possible to use carp-mola polyculture ponds where domesticated stocks are available as a reliable source of mola and SIS brood fish.

Hardening of Mola and SIS Broodfish for Live Transport

- ❖ The process of repeated netting and releasing of mola in ponds is necessary for proper hardening before transportation in live condition.
- ❖ This process needs to be repeated at least 2-3 times before transportation using soft seine net with continuous shower manually.
- ❖ Mola and SIS broodfishes should be fed with fine rice bran or powdered commercial feed for 3-5 days before harvesting and transportation.
- ❖ The hardening of mola is done similar to hardening of the carp fry/fingerlings – for which proper feeding and netting is required so that the brood fishes become stronger.

Mola and SIS broodfish collection and transportation from ponds

- ❖ Mola and SIS broodfish should be transported during low temperatures. Thus, early morning and late evening are most preferred for Mola and SIS broodfish transportation.
- ❖ Feeding must be stopped on the day prior to harvesting and transportation.
- ❖ Soft seine net should be used for collection of the broodfish. Large mesh screening net (Grader Net) can be used to separate the mola and SIS broodfish from large fish.
- ❖ Collected mola and SIS have to be transferred to a hapa set in the collection pond.

- ❖ Continuous shower of water must be given throughout the collection and conditioning process.
- ❖ Strong and healthy brood fishes can be identified with good swimming behaviour against water current and these are collected for transportation.
- ❖ During transportation, oral saline or salt @ 10 g/ 30-40 liter can be used for metabolic regulation in fish.
- ❖ Oxygen releasing agent (eg., calcium peroxide, sodium peroxide, sodium percarbonate) can be used to get better result.
- ❖ The whole process of netting and transportation should be done slowly and very carefully to avoid any injury that may result in secondary infection and mortality of the brood fishes.

Stocking

- ❖ Pond is ready to stock once water turns green (rich in plankton – natural food for fish)
- ❖ Purchase fish seeds from Department hatcheries, Multiplication centres or Registered Private hatcheries. Stocking of genetically improved Jayanti rohu and Amur carp is highly recommended.
- ❖ Treat the fish seeds with KMnO_4 @1-3 ppm or NaCl @4% before stocking.
- ❖ Maintain stocking density (8000 nos/ha) as recommended.
- ❖ Maintain stocking ratio of 4:3:3 (surface feeder: column feeder: bottom feeder).
- ❖ Stock Mola and SIS brood fish @ 50 kg/ha along with IMC

Water quality management

- ❖ Monitor the pond regularly (early morning and late in the afternoon):
- ❖ Low DO stress is likely:
 - If the pond is dark green with Secchi disc visibility of <15 cm (DO stress occurs either very early in the morning or when the day is cloudy)
 - If pond water is dark/black coloured with foul smell (eg, H_2S)
- ❖ Check pH after heavy rainfall and apply lime if pH is <6.5

Feeding

- ❖ Follow feeding options, depending on the feed availability and costs:
 - Floating pellets
 - Sinking pellets
 - Farm-made feed
 - Combination of commercial formulated feed and farm-made feed
- ❖ Adjust feeding rate as recommended

Regular Sampling

- ❖ Do sampling every month to assess the growth and health of the cultured fishes:
 - Monitor the length and weight of different species of fishes stocked.
 - Use the information for adjusting the feeding rates.
 - Check for the presence of external parasites on the body surface, fins, gills etc. and for the presence of lesions or wounds.

Disease Management

- ❖ There are a numbers of preventive measures farmers can use to reduce the risk of diseases and parasitic infection:
 - Stock pond with healthy, disease-free seed.
 - Disinfect all tools used in fish culture operations.
 - Avoid overcrowding.
 - Avoid restocking the ponds with fishes from unknown sources. If unavoidable, check the health of fishes stocked.
 - Follow proper feeding, fertilization and water quality management practices.
 - Conduct routine screening for any sign of disease/stress.
 - Record all clinical signs; take pictures of clinical signs, and record of number of dead fish.
 - Remove all moribund and dead fish and bury them far away from the pond.
 - Bring any unusual mortality to the attention of responsible authorities and assist them to collect sick and moribund samples for laboratory testing.
 - Follow the suggestion of the responsible authority for disease management

Harvesting

- ❖ Follow harvesting strategy based on culture system – generally single stocking and multiple harvesting is recommended:
 - Harvest fishes when they grow to 800g -1 kg. size.
 - Condition fishes for 4 to 6 hours before sending to live fish market

Harvesting of Mola & SIS

Partial harvesting of mola:

- ❖ Mola and SIS are continuous breeders and have a long breeding season (March - October).
- ❖ They attain maturity early and mostly have high fecundity. Therefore, partial harvesting of mola and SIS is very essential; otherwise, increasing density of mola and SIS will hamper the total Mola-SIS production.

- ❖ After 2-3 months of mola and SIS broodfish stocking, partial harvesting can be started at regular intervals of 15-30 days.
- ❖ Use large, meshed grader net to collect mola-SIS in live condition for marketing

Bulk harvesting of mola and SIS:

- ❖ According to production and market demand, bulk harvesting of mola and SIS can be done in two seasons depending on the size of the ponds.
- ❖ Once in November-December (before winter) and another is April-May (before monsoon).
- ❖ Large meshed Seine net (Grader net) is used for bulk harvesting of mola and SIS.
- ❖ Before bulk harvest, market demand and price should be analysed.

Regular harvesting of mola and SIS for household consumption:

- ❖ Mola and SIS collected from the carp-mola-SIS production system can be a good supplement to meet the nutritional requirement of the household.
- ❖ Gill net of 0.60-0.75cm mesh size can be used for regular harvesting of mola and SIS for household consumption.
- ❖ Cast net, lift net etc. can also be used for mola and SIS harvesting.

Record Keeping

- ❖ Maintain records of all farm activities, stocking details, inputs applied, sampling details, harvest details, sales and consumption.
- ❖ Use record books provided under the project for keeping records.
- ❖ Ensure the Project staff or Department officials update the records periodically during their visit to the farm.

Chapter 3



Polyculture of carps with genetically improved fish strains



3 Polyculture of carps with genetically improved fish strains

Introduction

Fish Polyculture is the system in which fast growing compatible species of different feeding habits and possibly of different age groups of the same species, are stocked together in the same water body (ponds, lakes etc) so that all its ecological niches are occupied by fish. Main target is to obtain high production from unit of waterbody. Fast growing compatible species of different feeding habit are stocked in different proportion in the same pond so that all its ecological niches are occupied. The maximum utilization of all the requirements of life takes place without doing any harm to each other. Polyculture of carps is widely practiced by the fish farmers in Assam.

Jayanti Rohu and Amur Common Carp are genetically improved strains with higher growth efficiency developed through selective breeding. Morphologically Jayanti Rohu is similar to normal Rohu but its growth rate is approximately 13-17 % higher than normal Rohu. Similarly, the growth of Amur common carp is approximately 27 % higher than normal Common carp and its market demand is high due to round body and slim belly compared to normal Common carp.

Pond Preparation

- ❖ Drain pond; remove sludge from the bottom and let it dry
- ❖ Ploughing of pond bottom
- ❖ Keep pond dyke clean from bushes to prevent hiding of predators
- ❖ Repair dike as necessary to control water seepage
- ❖ Repair inlet and outlet pipes
- ❖ Place bamboo or other substrates – which help in colonizing periphyton and also help in protecting from poaching of fish
- ❖ Apply lime, distributing evenly over the pond surface, at a recommended rate
- ❖ Consider fencing the sides of pond using nylon net/bamboo/any other locally available materials or bio-fencing using plants like Assam Lemon, Chitranaalla

Water Filling and fertilization of pond

- ❖ Fill pond; maintain water depth between 1.5-2.0 m. Filter water with double lined nylon nets if water from natural sources is used for filling.
- ❖ Fertilize pond, using chemical fertilizers @28 kg N and 21 kg P/ha with Urea and single super phosphate OR other fertilizers readily available in local market.

- ❖ Applying recommended doses of organic manure (Raw Cow Dung @10000 kg/ha) in heaps in one or two pond corners enclosed with the help of split bamboo.

Stocking

- ❖ Pond is ready to stock once water turns green (rich in plankton – natural food for fish)
- ❖ Purchase fish seeds from Department hatcheries, Multiplication centres or Registered Private hatcheries.
- ❖ Treat the fish seeds with KMnO_4 @1-3 ppm or NaCl @4% before stocking
- ❖ Maintain stocking density (8000 nos/ha) as recommended.
- ❖ Maintain stocking ratio of 4:3:3 (surface feeder: column feeder: bottom feeder).

Stocking of Genetically Improved Fish Strains

- ❖ Procure genetically improved fish seeds from Department hatcheries, Multiplication centres or Registered Private hatcheries having genuine stocks of Jayanti Rohu and Amur common carp.
- ❖ Replace the normal Rohu and Common Carp fish seeds with Jayanti Rohu and Amur Carp seeds.
- ❖ Treat the Jayanti Rohu and Amur Carp fish seed with KMnO_4 @1-3 ppm or NaCl @4% before stocking.
- ❖ Maintain stocking density as recommended for normal rohu and common carp.

Water quality management

- ❖ Monitor the pond regularly (early morning and late in the afternoon):
- ❖ Low DO stress is likely:
 - If the pond is dark green with Secchi disc visibility of <15 cm (DO stress occurs either very early in the morning or when the day is cloudy).
 - If pond water is dark/black coloured with foul smell (e.g., H_2S)
- ❖ Check pH after heavy rainfall and apply lime if pH is <6.5

Feeding

- ❖ Follow feeding options, depending on the feed availability and costs:
 - Floating pellets
 - Sinking pellets
 - Farm-made feed
 - Combination of commercial formulated feed and farm-made feed
- ❖ Adjust feeding rate as recommended

Regular Sampling

- ❖ Do sampling every month to assess the growth and health of the cultured fishes:

- Monitor the length and weight of different species of fishes stocked.
- Use the information for adjusting the feeding rates
- Check for the presence of external parasites on the body surface, fins, gills etc. and also for the presence of lesions or wounds.

Disease Management

Preventive measures farmers can use to reduce the risk of diseases and parasitic infection:

- ❖ Stock pond with healthy, disease-free seed.
- ❖ Disinfect all tools used in fish culture operations.
- ❖ Avoid overcrowding.
- ❖ Avoid restocking the ponds with fishes from unknown sources. If unavoidable, check the health of fishes stocked
- ❖ Follow proper feeding, fertilization and water quality management practices.
- ❖ Conduct routine screening for any sign of disease/stress.
- ❖ Record all clinical signs; take pictures of clinical signs, and record of number of dead fish.
- ❖ Remove all moribund and dead fish and bury them far away from the pond.
- ❖ Bring any unusual mortality to the attention of responsible authorities and assist them to collect sick and moribund samples for laboratory testing.
- ❖ Follow the suggestion of the responsible authority for disease management

Harvesting

- ❖ Follow harvesting strategy based on culture system – generally single stocking and multiple harvesting is recommended:
 - Harvest fishes when they grow to 800g -1 kg. size.
 - Condition fishes for 4 to 6 hrs before sending to live fish market

Record Keeping

- ❖ Maintain records of all farm activities, stocking details, inputs applied, sampling details, harvest details, sales and consumption.
- ❖ Use record books provided under the project for keeping records.
- ❖ Ensure the Project staff or Department officials update the records periodically during their visit to the farm.

Chapter 4



Polyculture of carps with freshwater prawn



4 Polyculture of carps with freshwater prawn

Introduction

Giant freshwater prawn *Macrobrachium rosenbergii* is commonly known as “SCAMPI”. It is widely distributed in Southeast Asia and found in most river systems in India. It is highly valued candidate species for aquaculture due to its high price, large size, rapid growth, good taste and high export demand. It can also be cultured in freshwater as well as slightly brackish water.

Polyculture of giant freshwater prawn along with Indian major carps is becoming popular compared to monoculture among fish farmers. The presence of fish in a polyculture system serves as a biological control over development of zooplankton, phytoplankton and filamentous algae, which otherwise result in ecological instability of the pond ecosystem. Since the prawn under polyculture with fish derives its nutrition by utilizing the natural pond productivity and leftover fish feed, faecal matter of fish, etc., there is no need for a separate high-cost prawn feed. The inclusion of freshwater prawns in a polyculture system usually has synergistic beneficial effects, which include:

- ✓ More stable dissolved oxygen levels,
- ✓ The reduction of predators,
- ✓ Increased pond productivity (all species),
- ✓ The potential to increase the total value of the crop by the inclusion of a high-value species.

Pond Preparation

- ❖ Drain pond; remove sludge from the bottom and let it dry
- ❖ Keep pond dyke clean from bushes to prevent hiding of predators
- ❖ Repair dyke as necessary to control water seepage
- ❖ Repair inlet and outlet pipes
- ❖ Place bamboo or other substrates – which help in colonizing periphyton and also help in protecting from poaching of fish
- ❖ Apply lime, distributing evenly over the pond surface, at a recommended rate
- ❖ Consider fencing the sides of pond using nylon net/bamboo/any other locally available materials or bio-fencing using plants like Assam Lemon, Chitranaalla
- ❖ Drain annually OR once in two years – depending on the need
- ❖ Liming is must if soil pH in the pond is <6.5
- ❖ Consider using cost-effective materials for fencing or provide bio-fencing

Water Filling and fertilization of pond

- ❖ Fill pond, maintain water depth between 1.5-2.0 m. Filter water with double lined nylon nets if water from natural sources is used for filling.
- ❖ Fertilize pond, using chemical fertilizers @ 28 kg N and 21 kg P/ha with Urea and single super phosphate OR other fertilizers readily available in local market.
- ❖ Consider applying recommended doses of organic manure (livestock manure/chicken manure/compost) in heaps in one or two pond corners enclosed with the help of split bamboo. Consider application of bio-fertilizers like Anabaena – Azolla complex
- ❖ Install a water level measuring pole marked at 1m, 1.5m and 2 m to check water level
- ❖ Since prawns are cannibalistic in nature and having territorial behaviour, so they necessitate special hiding arrangement during moulting phase. It is to be noted that moulting is the only process of growth of shellfish.
- ❖ During moulting phase, prawns are weak and are exposed with soft muscle. Thus to prevent the cannibalistic nature during moulting plastic pipe, hollow bamboo, beetalnut leaves, coconut leaves, tyres, tree branches, etc. need be provided as hiding arrangement.

Stocking

- ❖ Pond is ready to stock once water turns green (rich in plankton – natural food for fish)
- ❖ Acquire fish seed from Department hatcheries, Multiplication centres or Registered Private hatcheries (where available, acquire improved Jayanti Rohu)
- ❖ Procure fish seeds of 10 cm size for stocking. If fingerlings are small (<5 cm), nurse them in nursery pond until they grow to 10 cm; feed the fingerlings with high- quality feed
- ❖ Stock Fresh water prawn juveniles of size 4-6 cm and maintained stocking density (7500 nos/ha) as recommended
- ❖ Treat the seeds with KMnO_4 @ 1-3 ppm or NaCl @ 4% before stocking
- ❖ Maintain fish stocking density (2500 - 3500 nos/ha) as recommended
- ❖ Maintain stocking ratio of 5:5 (surface feeder: column feeder)
- ❖ Avoid stocking fish seeds on weight basis and follow stocking recommended species in the recommended ratio and density
- ❖ Adjust stocking density based on harvesting strategy (depending on whether fish to be harvested as 'hotel size fish' or to be grown >1 kg size; single stocking multiple harvesting)

Crop Calendar

- ❖ Follow the crop calendar given below to plan fish production to reach the market during the ban period in Assam *i.e.*, April-June when fish demand and fish prices are high
- ❖ Ensure stocking during July-August so that the fishes grow to marketable size by March
- ❖ Fingerlings of different species should be made available for stocking in August.
- ❖ Advanced nursing on-farm reduce seed cost substantially and enable availability of required seed on-farm for stocking.
- ❖ Early season stocking enables farmers to harvest fish in April to June when fish supply in market is low and fetch better price.
- ❖ Short production cycle (March – May) for production of hotel size fish can be followed in the areas prone to flood damage.

Months	July	August to March	April	May	June
Activities	Pond preparation & Stocking	Growing	Harvesting		

Water quality management

- ❖ **Low DO stress is likely:**
 - If the pond is dark green with Secchi disc visibility of <15 cm (DO stress occurs either very early in the morning or when the day is cloudy)
 - If pond water is dark/black coloured with foul smell (e.g., H₂S)
 - Check pH after heavy rainfall and apply lime if pH is <6.5
- ❖ **Low DO stress symptom:**
 - Fish start gasping for air on pond surface OR sometimes even die if DO is too low.
 - If such symptoms are noticed, partially exchange pond water or sprinkle water over the pond to increase DO level.
- ❖ Mechanical aerators for large farms; for small ponds pumping water and creating like fountain over the pond can be done.
- ❖ If pond watercolour turns black/dark, drain some bottom water and top up with new water.

Feeding

- ❖ Follow feeding options, depending on the feed availability and costs:
 - Sinking pellets
 - Farm-made feed
 - Combination of commercial formulated feed and farm-made feed

- ❖ Adjust feeding rate as recommended
- ❖ Ideal to have 30% CP in grow out feed:
- ❖ broadcast feed to pond from the surround dykes
- ❖ place feed in feeding trays (made of locally available materials) submerged in the pond
- ❖ put feed in perforated bags and submerge just below the water surface so that fish can browse feed and where feed loss is minimal
- ❖ Feed can also be given in check trays placed 2-3 m away from the dyke for better feed management
- ❖ Adjust feeding based on fish biomass (feeding table to be provided to the farmer)
- ❖ It is advised to cook farm made feed to make it more palatable.
- ❖ Daily feeding rate range from 5% of total fish biomass (for fry/fingerlings) and 1.0% for large size fish
- ❖ Feeding rate also needs to be adjusted based on feed quality, plankton richness in the pond, etc.
- ❖ Avoid feeding around mid-day/when temperature is very high/during the cloudy/rainy days

Regular Sampling

- ❖ Do sampling every month to assess the growth and health of the cultured fishes and prawns:
- ❖ Monitor the length and weight of different species of fishes stocked.
- ❖ Monitor the length and weight of cultured prawns.
- ❖ Use the information for adjusting the feeding rates
- ❖ Check for the presence of external parasites on the body surface, fins, gills etc. and for the presence of lesions or wounds.

Disease Management

- ❖ Stock pond with healthy, disease-free seed.
- ❖ Disinfect all tools used in fish culture operations.
- ❖ Avoid overcrowding.
- ❖ Follow proper feeding, fertilization and water quality management practices.
- ❖ Conduct routine screening for any sign of disease/stress.
- ❖ Record all clinical signs; take pictures of clinical signs, and record of number of dead fish.
- ❖ Remove all moribund and dead fish and bury them far away from the pond.
- ❖ Bring any unusual mortality to the attention of responsible authorities and assist them to collect sick and moribund samples for laboratory testing.
- ❖ Follow the suggestion of the responsible authority for disease management.

Harvesting

- ❖ Harvest fishes when they grow to 800g -1 kg. size
- ❖ Harvest prawns when they grow more than 50 g in size.
- ❖ Condition fish before sending to live fish market
- ❖ Harvesting strategy largely depends on purpose, market preference, growout system, etc.
- ❖ Harvest fish 1 day before sending to market and put them in nylon hapa; stop feeding at least 24 hours before sending to market
- ❖ Large prawns (>50 g) may be harvested using seine net of suitable mesh size after four months of culture, which should continue once every 3-4 weeks thereafter for the next 3-4 months. The prawns may be finally harvested after 8 months of culture by complete dewatering.

Marketing

- ❖ Send fish to market either early in the morning or late in the afternoon
- ❖ Preferably send fishes live to the market to get better price
- ❖ Avoid transportation of fish when temperature is high (mid-day to early afternoon)
- ❖ While transporting live fish maintain biomass based on volume of water to avoid mortality

Environmental Care

- ❖ Avoid draining pond water and sludge directly to nearby water bodies
- ❖ Properly dispose dead fish (they should be removed immediately from the pond and buried properly)
- ❖ Properly dispose the feed bags, fertilizer bags, used bottles and containers

Record Keeping

- ❖ Maintain records of all farm activities, stocking details, inputs, sampling details, harvest details, selling and consumption.
- ❖ Use record books provided under the project for keeping records
- ❖ Ensure the Project staff or Department officials update the records periodically during their visit to the farm.

Chapter 5



Paddy cum fish integrated farming



5 Paddy cum fish integrated farming

Introduction

The paddy-cum-fish culture is an innovative farming system in which paddy is the main crop and fish is taken as an additional crop to earn extra income in the same farming area. The production of fish crop along with paddy crop gives the farmer an off-season occupation, which increases the income without increasing much expense. Apart from the additional income from paddy cum fish culture, leads to a reduction in labour in weeding and an increase in the yield of paddy by 5 to 15%. In paddy cum fish integrated farming systems, paddy and fish forms a mutualistic symbiosis as

- Increase in organic fertilization by fish excreta and remains of feed.
- Reduction in the number of harmful insects, such as paddy stem borers, whose larvae eaten by fish.
- Reduction in rat population due to increased water level.
- Increased mineralization of the organic matter and increased aeration of the soil resulting from the puddling of mud by benthic feeders.
- Control of algae and weeds that compete with rice for light and nutrients.

Paddy varieties

- ❖ BINA Dhan 11, Culture -1, Rangadoria, Fapori, Govind, Lachit and CH-63.

Paddy Seed quality

- ❖ Seed should be clean (contains no stones, soil particles, weed seeds), genetically pure (contains grains of only one variety) and healthy (full big grains, same colour, no cracks, no obvious disease or pest attack).
- ❖ Procure quality seeds of recommended paddy varieties through ATMA/KVK, Department of Agriculture/AAU/IRRI approved seed suppliers

Paddy Seed selection

- ❖ Seeds are immersed in clean water and stirred well.
- ❖ Seeds those sinks are selected, and those that float are rejected.
- ❖ Seeds can also be treated in brine solution (20%), washed and dried in shade.

Paddy Seed treatment

- ❖ Once the selection process is done, the seeds should be soaked directly in one of the following fungicidal suspensions for 12 hours.
- ❖ One litre of fungicidal solution is required to treat one kg of seed.
- ❖ Treated seeds should be kept in incubation for 24-36 hours before sowing in nursery beds
- ❖ Treat the seed with *Trichoderma harzianum* @ 10 g /kg seed overnight or 6 hrs before sowing.
- ❖ Dry the seed in shade for 30 minutes after seed treatment.

Nursery Raising

- ❖ **Time of sowing:** January-February season is the most appropriate time for raising nursery for targeting the transplanting after 25-30 days when they are at 5-6 leaf stage.
- ❖ **Seed rate:** 40-45 kg per hectare
- ❖ **Nursery-bed size:**
 - For one hectare of main field, the nursery seedbed size should be between 750 and 1000 m² and not more than 1/10th of the main field size.
 - Land is thoroughly puddled and seedbeds of 10m length and 1.25m breadth, 15 cm height are prepared with 50 cm gap in between the beds.
 - The length of the bed may vary according to convenience and availability of space.
- ❖ **Irrigation:** Irrigate as per the requirement to maintain water saturated condition of the seedbed. Maintain 2-3 cm standing water for 2-3 days prior to uprooting.
- ❖ **Nursery fertilizer:** Use 3-1-2g N-P₂O₅-K₂O + 2000g manure per m² that is equivalent to 3-1-2 kg N-P₂O₅-K₂O + 2 tonnes manure for 1000 m² of nursery area.

Table 1: Fertilizer use in nursery

When?	What?	How much? g/m ² or kg/1000 m ²		Application method
		Through DAP	Through SSP	
Basal (At sowing)	Manure	2000	2000	Soil incorporation
	Urea	5.6	6.5	
	DAP	2.2	-	
	SSP	-	6.3	
	MOP	3.3	3.3	

- ❖ **Plant protection:** Adopt plant protection measures in the standing nursery as per standard package of practices
- ❖ **Seedling priming against insects:** As protective measure against stem borer, gall midge and hoppers, the uprooted seedlings are first washed and then root portion is dipped into 0.2% solution of Chlorpyrifos (2 ml/lit of water) along with 1% urea (10g/lit of water) for 3 hours. Alternatively, in standing nursery 5-7 days before uprooting, apply or spray Chlorpyrifos 20 EC @ 0.2%.

Trench/pond preparation

- ❖ Create a pond or ditch for 30% to 40% of the main field area with a depth of 3-4 feet at the lowest side of the paddy plot, and raise the dyke to a minimum of 2 feet above the local flood height at peak rainy season
- ❖ Renovate the existing canals, pond and trenches to retain water for the fishes during the summer months
- ❖ Install inlet and outlet pipes for maintenance of water level during the culture period.
- ❖ The inlets and outlets should be guarded by bamboo split or wire/plastic mesh (*Bana*) to prevent weed fish and aquatic predators from entering into paddy fields and to prevent escape of the stocked fishes.
- ❖ Liming should be done in the pond/trench area as well as in the canals of paddy field with Quicklime at a recommended rate of 80 kg/ha

Main Paddy field Preparation

- ❖ Field should be prepared thoroughly by ploughing 4 to 5 times, followed by harrowing and laddering. Ploughing should be started at least 21 days before transplanting so that weeds are dried up/decomposed. Alternatively, one pass of mouldboard plough followed by one or two passes of modified helical blade puddler to be done for obtaining good quality puddled soil.
- ❖ The ploughing intervals should be spaced such that the weeds germinating after the first round of ploughing are ploughed up in the next round.
- ❖ If available, apply compost or well decomposed farmyard manure uniformly prior to field preparation and mix it well with soil.
- ❖ Repair the bunds to prevent water leakage from the field during the cropping season.
- ❖ Field should be puddled and levelled well, better maintaining a thin and uniform layer of water
- ❖ The basal fertilizer dose is applied to the field as per recommendation in the Table below:

Table 2: Fertilizer use in main field

Stage of fertiliser application	Name of fertiliser s	Fertiliser application				Application Method
		(kg/ha)		(kg/bigha)		
		<i>Boro/Ahu</i> season	<i>Sali</i> season	<i>Boro/Ahu</i> season	<i>Sali</i> season	
Basal Same day at transplanting	Compost or FYM	10000	10000	1300	1300	Broadcast & incorporate in soil at the time of filed preparation
	Urea	14.6	22	2.0	3.0	
	SSP	62	62	8.3	8.3	
	MOP	16	16	2.1	2.1	

Transplanting

- ❖ **Seedling age:** Always use 25 to 30 days old (5-6 leaf stage) seedlings for transplanting for better yield
- ❖ **Spacing:** Transplant at distance of 20 x15 cm (8x6 inches)
- ❖ **Seedling density:** 2-3 seedlings per hill
- ❖ **Depth of seedling:** 4-5 cm

Fish stocking

- ❖ After 15-20 days of paddy transplantation, stocking of fishes can be done.
- ❖ Procure quality fish seeds from department hatcheries, multiplication centres or Registered Private hatcheries (where available improved Jayanti Rohu, Amur carp seeds)
- ❖ Procure fish seeds of 10 cm size for stocking. If fingerlings available in local hatcheries are small (<5 cm), procure in advance and nurse them in nursery pond until they grow to 10 cm;
- ❖ Treat the fish seed with KMnO₄ @1-3 ppm or NaCl @4% before stocking
- ❖ Maintain stocking density (10000 nos/ha) as recommended
- ❖ The hardening of mola is done similar to hardening of the carp fry/fingerlings – for which proper feeding and netting is required so that the brood fishes become stronger.

Fertilizer management for paddy

- ❖ For transplanted rice, fertilizer recommendation per hectare is as per Assam Agricultural University recommendations for paddy-fish farming.
- ❖ The nitrogen dose is split into 3 equal applications - about 1/3 as basal, 1/3 at tillering and 1/3 at panicle initiation.

Water quality management for fish culture

Low Dissolved Oxygen stress symptoms:

- Fish come to surface and gasping for air
- Shows lethargic behavior
- Stop eating feed

Low Dissolved Oxygen stress is likely:

- If the pond is dark green with Secchi disc visibility of <15 cm (DO stress occurs either very early in the morning or when the day is cloudy)
- If pond water is dark/black coloured with foul smell (e.g., H₂S)
- Low Dissolved Oxygen stress occurs either early in the morning or when the day is cloudy
- ❖ Check pH after heavy rainfall and apply quick lime if pH is <6.5 at the rate of 200 kg/ha

Weeding: Keep the field weed-free especially in the early stage of crop as weeds do most damage in early crop stage. However, later control is also important to prevent seed setting by weeds.

Hand weeding: 1st weeding 3 weeks after transplanting and 2nd weeding 6 weeks after transplanting

Feeding the fishes

- ❖ Follow feeding options, depending on the feed availability and costs:
 - Floating pellets
 - Sinking pellets
 - Farm-made feed
 - Combination of commercial formulated feed and farm-made feed
- ❖ Adjust feeding rate as recommended

Plant protection measures

- ❖ Follow plant protection measures as per the recommendations. However, no pesticides/herbicides to be applied in in paddy –fish farming system as it will result in mortality of stocked fishes.

Regular sampling of fishes

- ❖ Monitor the length and weight of different species of fishes stocked.
- ❖ Use the information for adjusting the feeding rates
- ❖ Check for the presence of external parasites on the body surface, fins, gills etc. and for the presence of lesions or wounds.

- ❖ Adjust feeding rate based on total fish biomass in the system: $(\text{fish biomass (kg)} = \text{average size of fish (g)} \times \text{number of fish stocked (n)})/1000$.

Disease management of fishes

Unlike pond culture, incidence of fish disease is low in paddy –fish system. There are a numbers of preventive measures farmers can follow to reduce the risk of diseases and parasitic infection:

- ❖ Stock pond with healthy, disease-free seed.
- ❖ Disinfect all tools used in fish culture operations.
- ❖ Avoid overcrowding.
- ❖ Follow proper feeding, fertilization and water quality management practices.
- ❖ Conduct routine screening for any sign of disease/stress.
- ❖ Record all clinical signs; take pictures of clinical signs, and record of number of dead fish.
- ❖ Remove all moribund and dead fish, and bury them far away from the pond.
- ❖ Bring any unusual mortality to the attention of responsible authorities and assist them to collect sick and moribund samples for laboratory testing.

Harvest and post-harvest of paddy

- ❖ Harvest when 80-85% of the grains attain physiological maturity *i.e.*, visually straw coloured
- ❖ Minimize the time during which the harvested plants remain in the field and avoid field drying. Make sure that the panicles stay dry.
- ❖ Thresh and dry within two days after harvesting. Where available, adopt mechanised drying using solar bubble dryer. In the absence of mechanised drying, alternatively follow Sun drying on a mat or plastic sheet, keeping the thickness of the grain layer 3 to 5 cm.
- ❖ Clean the grain thoroughly by winnowing. Store the rice in cool, dry and clean storage bins.

Fish harvesting

- ❖ Generally single stocking multiple harvesting is recommended
- ❖ Partial harvest after 5 months to thin the stock by harvesting the bigger fishes
- ❖ Harvest fishes when they grow to marketable size
- ❖ Condition fishes overnight in holding tanks with sufficient aeration and water circulation before sending live to market to reduce mortality

Marketing of fish

- ❖ Transport fish to market either early in the morning or late in the afternoon
- ❖ Preferably send fishes live to the market to get better price

- ❖ For distant markets, pack fish with adequate ice and transport in insulated trucks.

Environmental care

- ❖ Make sure that fish farming activities are not causing damage to surrounding aquatic and terrestrial environment.
 - Avoid draining pond water and sludge directly to nearby water bodies
 - Properly dispose dead fish (they should be removed immediately from the pond and buried properly)
 - Properly dispose the feed bags, fertilizer bags, used bottles and containers

Record keeping

- ❖ Maintain records of all farm activities, stocking details, inputs, sampling details, harvest details, sale and consumption.
- ❖ Use record books provided under the project for keeping records.
- ❖ Ensure the records are updated periodically with the support of Project staff or Department officials during their visit.

Paddy-fish farming calendar

Follow the farming calendar presented below:

Months	Activities	
	Paddy farming	Fish farming
January	Sowing, Nursery raising and Transplanting	Civil works
February		Pond Preparation
March		Fish Seed Stocking
April		
May	Harvesting	
June		
July		Partial Harvesting
August		
September		
October		Fish Harvesting
November		
December		

Chapter 6



Paddy cum fish integrated farming with mola & Small Indigenous Fish Species (SIS)



6 Paddy cum fish integrated farming with mola & Small Indigenous Fish Species (SIS)

Introduction

The paddy-cum-fish culture is an innovative farming system in which paddy is the main crop and fish is taken as an additional crop to earn extra income in the same farming area. The production of fish crop along with paddy crop gives the farmer an off-season occupation, which increases the income without increasing much expense. Apart from the additional income from paddy cum fish culture, leads to a reduction in labour in weeding and an increase in the yield of paddy by 5 to 15%. In paddy cum fish integrated farming systems, paddy and fish forms a mutualistic symbiosis as

- Increase in organic fertilization by fish excreta and remains of feed.
- Reduction in the number of harmful insects, such as paddy stem borers, whose larvae eaten by fish.
- Reduction in rat population due to increased water level.
- Increased mineralization of the organic matter and increased aeration of the soil resulting from the puddling of mud by benthic feeders.
- Control of algae and weeds that compete with rice for light and nutrients.

Assam has abundant resources of small indigenous fish species (SIS) rich in micronutrients such as vitamin A, calcium, Iron, zinc etc. Among the several species of SIS, *Amblypharyngodon mola* locally called *Mola* is a nutrient dense fish, it contains vitamin-A, Iron, Zinc, and Calcium and these nutrients are highly required for pregnant and lactating women as well as young children. As mola is a potential candidate species and can be cultured along with carps in paddy fish farming without hampering the target species. The mola and SIS production technology can provide an additional production and income to the farmers and regular consumption of mola and SIS help to eradicate malnutrition and provide nutritional security for the local communities.

Paddy varieties

- ❖ BINA Dhan 11, Culture -1, Rangdoria, Fapori, Govind, Lachit and CH-63.

Paddy Seed quality

- ❖ Seed should be clean (contains no stones, soil particles, weed seeds), genetically pure (contains grains of only one variety) and healthy (full big grains, same colour, no cracks, no obvious disease or pest attack).
- ❖ Procure quality seeds of recommended paddy varieties through ATMA/KVK, Department of Agriculture/AAU/IRRI approved seed suppliers

Paddy Seed selection

- ❖ Seeds are immersed in clean water and stirred well.
- ❖ Seeds those sink are selected, and those that float are rejected.
- ❖ Seeds can also be treated in brine solution (20%), washed and dried in shade.

Paddy Seed treatment

- ❖ Once the selection process is done, the seeds should be soaked directly in one of the following fungicidal suspensions for 12 hours.
- ❖ One litre of fungicidal solution is required to treat one kg of seed.
- ❖ Treated seeds should be kept in incubation for 24-36 hours before sowing in nursery beds
- ❖ Treat the seed with *Trichoderma harzianum* @ 10 g /kg seed overnight or 6 hrs before sowing.
- ❖ Dry the seed in shade for 30 minutes after seed treatment.

Nursery Raising

- ❖ **Time of sowing:** January-February season is the most appropriate time for raising nursery for targeting the transplanting after 25-30 days when they are at 5-6 leaf stage.
- ❖ **Seed rate:** 40-45 kg per hectare
- ❖ **Nursery-bed size:**
 - For one hectare of main field, the nursery seedbed size should be between 750 and 1000 m² and not more than 1/10th of the main field size.
 - Land is thoroughly puddled and seedbeds of 10m length and 1.25m breadth, 15 cm height are prepared with 50 cm gap in between the beds.
 - The length of the bed may vary according to convenience and availability of space.
- ❖ **Irrigation:** Irrigate as per the requirement to maintain water saturated condition of the seedbed. Maintain 2-3 cm standing water for 2-3 days prior to uprooting.
- ❖ **Nursery fertilizer:** Use 3-1-2g N-P₂O₅-K₂O + 2000g manure per m² that is equivalent to 3-1-2 kg N-P₂O₅-K₂O + 2 tonnes manure for 1000 m² of nursery area.

Table 3: Fertilizer use in nursery

When?	What?	How much? g/m ² or kg/1000 m ²		Application method
		Through DAP	Through SSP	
Basal (At sowing)	Manure	2000	2000	Soil incorporation
	Urea	5.6	6.5	
	DAP	2.2	-	
	SSP	-	6.3	
	MOP	3.3	3.3	

- ❖ **Plant protection:** Adopt plant protection measures in the standing nursery as per standard package of practices
- ❖ **Seedling priming against insects:** As protective measure against stem borer, gall midge and hoppers, the uprooted seedlings are first washed and then root portion is dipped into 0.2% solution of Chlorpyrifos (2 ml/lit of water) along with 1% urea (10g/lit of water) for 3 hours. Alternatively, in standing nursery 5-7 days before uprooting, apply or spray Chlorpyrifos 20 EC @ 0.2%.

Trench/pond preparation

- ❖ Create a pond or ditch for 30% to 40% of the main field area with a depth of 3-4 feet at the lowest side of the paddy plot, and raise the dyke to a minimum of 2 feet above the local flood height at peak rainy season
- ❖ Renovate the existing canals, pond and trenches to retain water for the fishes during the summer months
- ❖ Install inlet and outlet pipes for maintenance of water level during the culture period.
- ❖ The inlets and outlets should be guarded by bamboo split or wire/plastic mesh (*Bana*) to prevent weed fish and aquatic predators from entering into paddy fields and to prevent escape of the stocked fishes.
- ❖ Liming should be done in the pond/trench area as well as in the canals of paddy field with Quicklime at a recommended rate of 80 kg/ha

Main Paddy field Preparation:

- ❖ Field should be prepared thoroughly by ploughing 4 to 5 times, followed by harrowing and laddering. Ploughing should be started at least 21 days before transplanting so that weeds are dried up/decomposed. Alternatively, one pass of mouldboard plough followed by one or two passes of modified helical blade puddler to be done for obtaining good quality puddled soil.

- ❖ The ploughing intervals should be spaced such that the weeds germinating after the first round of ploughing are ploughed up in the next round.
- ❖ If available, apply compost or well decomposed farmyard manure uniformly prior to field preparation and mix it well with soil.
- ❖ Repair the bunds to prevent water leakage from the field during the cropping season.
- ❖ Field should be puddled and levelled well, better maintaining a thin and uniform layer of water
- ❖ The basal fertilizer dose is applied to the field as per recommendation in the Table below:

Table 4: Fertilizer use in main field

Stage of fertiliser application	Name of fertilisers	Fertiliser application				Application Method
		(kg/ha)		(kg/bigha)		
		<i>Boro/Ahu season</i>	<i>Sali season</i>	<i>Boro/Ahu season</i>	<i>Sali season</i>	
Basal Same day at transplanting	Compost or FYM	10000	10000	1300	1300	Broadcast & incorporate in soil at the time of filed preparation
	Urea	14.6	22	2.0	3.0	
	SSP	62	62	8.3	8.3	
	MOP	16	16	2.1	2.1	

Transplanting

- ❖ **Seedling age:** Always use 25 to 30 days old (5-6 leaf stage) seedlings for transplanting for better yield
- ❖ **Spacing:** Transplant at distance of 20 x15 cm (8x6 inches)
- ❖ **Seedling density:** 2-3 seedlings per hill
- ❖ **Depth of seedling:** 4-5 cm

Fish stocking

- ❖ After 15-20 days of paddy transplantation, stocking of fishes can be done.
- ❖ Procure quality fish seeds from department hatcheries, multiplication centres or Registered Private hatcheries (where available improved Jayanti Rohu, Amur carp seeds)
- ❖ Procure fish seeds of 10 cm size for stocking. If fingerlings available in local hatcheries are small (<5 cm), procure in advance and nurse them in nursery pond until they grow to 10 cm;
- ❖ Treat the fish seed with KMnO₄ @1-3 ppm or NaCl @4% before stocking
- ❖ Maintain stocking density (10000 nos/ha) as recommended

Identification of Mola and SIS broodfish source

- ❖ Mola and SIS are self- recruiting fish species inhabiting rivers, streams, beels, ponds, ditches, tanks and inundated paddy fields. It is commonly found in larger ponds connected to open rice fields and ditches in the floodplains.
- ❖ The mola and other SIS brood source need to be initially identified. Ideally, fish farming clusters should have good records of mola and SIS source that can be used for collection of matured broodfish. It is also possible to use carp-mola polyculture ponds where domesticated stocks are available as a reliable source of mola and SIS brood fish.

Hardening of Mola and SIS Broodfish for Live Transport

- ❖ The process of repeated netting and releasing of mola in ponds is necessary for proper hardening before transportation in live condition.
- ❖ This process need to be repeated at least 2-3 times before transportation using soft seine net with continuous shower manually.
- ❖ Mola and SIS broodfishes should be fed with fine rice bran or powdered commercial feed for 3-5 days before harvesting and transportation.
- ❖ The hardening of mola is done similar to hardening of the carp fry/fingerlings – for which proper feeding and netting is required so that the brood fishes become stronger.

Mola and SIS broodfish collection and transportation from ponds

- ❖ Mola and SIS broodfish should be transported during low temperatures. Thus, early morning and late evening are most preferred for Mola and SIS broodfish transportation.
- ❖ Feeding must be stopped on the day prior to harvesting and transportaion.
- ❖ Soft seine net should be used for collection of the broodfish. Large mesh screening net (Grader Net) can be used to separate the mola and SIS broodfish from large fish.
- ❖ Collected mola and SIS have to be transfered to a hapa set in the collection pond.
- ❖ Continuous shower of water must be given throughout the collection and conditioning process.
- ❖ Strong and healthy brood fishes can be identified with good swimming behaviour against water current and these are collected for transportation.
- ❖ During transportation, oral saline or salt @ 10 g/ 30-40 liter can be used for metabolic regulation in fish.
- ❖ Oxygen releasing agent (eg., calcium peroxide, sodium peroxide, sodium percarbonate) can be used to get better result.

- ❖ The whole process of netting and transportation should be done slowly and very carefully to avoid any injury that may result in secondary infection and mortality of the brood fishes.

Stocking of Mola- SIS

- ❖ Stock Mola and SIS brood fish @ 25 kg/ha along with IMC

Fertilizer management for paddy

- ❖ For transplanted rice, fertilizer recommendation per hectare is as per Assam Agricultural University recommendations for paddy-fish farming.
- ❖ The nitrogen dose is split into three equal applications - about 1/3 as basal, 1/3 at tillering and 1/3 at panicle initiation.

Water quality management for fish culture

Low Dissolved Oxygen stress symptoms:

- Fish come to surface and gasping for air
- Shows lethargic behavior
- Stop eating feed

Low Dissolved Oxygen stress is likely:

- If the pond is dark green with Secchi disc visibility of <15 cm (DO stress occurs either very early in the morning or when the day is cloudy)
- If pond water is dark/black coloured with foul smell (eg, H₂S)
- Low Dissolved Oxygen stress occurs either early in the morning or when the day is cloudy
- ❖ Check pH after heavy rainfall and apply quick lime if pH is <6.5 at the rate of 200 kg/ha

Weeding: Keep the field weed-free especially in the early stage of crop as weeds do most damage in early crop stage. However, later control is also important to prevent seed setting by weeds.

Hand weeding: 1st weeding 3 weeks after transplanting and 2nd weeding 6 weeks after transplanting.

Feeding the fishes

- ❖ Follow feeding options, depending on the feed availability and costs:
 - Floating pellets
 - Sinking pellets
 - Farm-made feed
 - Combination of commercial formulated feed and farm-made feed

- ❖ Adjust feeding rate as recommended

Plant protection measures

- ❖ Follow plant protection measures as per the recommendations. However, no pesticides/herbicides to be applied in in paddy –fish farming system as it will result in mortality of stocked fishes.

Regular sampling of fishes

- ❖ Monitor the length and weight of different species of fishes stocked.
- ❖ Use the information for adjusting the feeding rates
- ❖ Check for the presence of external parasites on the body surface, fins, gills etc. and for the presence of lesions or wounds.
- ❖ Adjust feeding rate based on total fish biomass in the system: $(\text{fish biomass (kg)} = \text{average size of fish (g)} \times \text{to number of fish stocked (n)})/1000$.

Disease management of fishes

Unlike pond culture, incidence of fish disease is low in paddy –fish system. There are a numbers of preventive measures farmers can follow to reduce the risk of diseases and parasitic infection:

- ❖ Stock pond with healthy, disease-free seed.
- ❖ Disinfect all tools used in fish culture operations.
- ❖ Avoid overcrowding.
- ❖ Follow proper feeding, fertilization and water quality management practices.
- ❖ Conduct routine screening for any sign of disease/stress.
- ❖ Record all clinical signs; take pictures of clinical signs, and record of number of dead fish.
- ❖ Remove all moribund and dead fish, and bury them far away from the pond.
- ❖ Bring any unusual mortality to the attention of responsible authorities and assist them to collect sick and moribund samples for laboratory testing.

Harvest and post-harvest of paddy

- ❖ Harvest when 80-85% of the grains attain physiological maturity i.e. visually straw coloured
- ❖ Minimize the time during which the harvested plants remain in the field and avoid field drying. Make sure that the panicles stay dry.
- ❖ Thresh and dry within two days after harvesting. Where available, adopt mechanised drying using solar bubble dryer. In the absence of mechanised drying, alternatively follow Sun drying on a mat or plastic sheet, keeping the thickness of the grain layer 3 to 5 cm.
- ❖ Clean the grain thoroughly by winnowing. Store the rice in cool, dry and clean

storage bins.

Fish harvesting

- ❖ Generally single stocking multiple harvesting is recommended
- ❖ Partial harvest after 5 months to thin the stock by harvesting the bigger fishes
- ❖ Harvest fishes when they grow to marketable size
- ❖ Condition fishes overnight in holding tanks with sufficient aeration and water circulation before sending live to market to reduce mortality

Partial harvesting of mola

- ❖ Mola and SIS are continuous breeders and have a long breeding season (March - October).
- ❖ They attain maturity early and mostly have high fecundity. Therefore, partial harvesting of mola and SIS is very essential; otherwise, increasing density of mola and SIS will hamper the total Mola-SIS production.
- ❖ After 2-3 months of mola and SIS broodfish stocking, partial harvesting can be started at regular intervals of 15-30 days.
- ❖ Use large meshed grader net to collect mola-SIS in live condition for marketing

Bulk harvesting of mola and SIS

- ❖ According to production and market demand, bulk harvesting of mola and SIS can be done in two seasons depending on the size of the ponds.
- ❖ Once in November-December (before winter) and another is April-May (before monsoon).
- ❖ Large meshed Seine net (Grader net) is used for bulk harvesting of mola and SIS.
- ❖ Before bulk harvest, market demand and price should be analysed.

Regular harvesting of mola and SIS for household consumption

- ❖ Mola and SIS collected from the carp-mola-SIS production system can be a good supplement to meet the nutritional requirement of the household.
- ❖ Gill net of 0.60-0.75cm mesh size can be used for regular harvesting of mola and SIS for household consumption.
- ❖ Cast net, lift net etc. can also be used for mola and SIS harvesting.

Marketing of fish

- ❖ Transport fish to market either early in the morning or late in the afternoon
- ❖ Preferably send fishes live to the market to get better price
- ❖ For distant markets, pack fish with adequate ice and transport in insulated trucks

Environmental care

- ❖ Make sure that fish farming activities are not causing damage to surrounding aquatic and terrestrial environment.
 - Avoid draining pond water and sludge directly to nearby water bodies
 - Properly dispose dead fish (they should be removed immediately from the pond and buried properly)
 - Properly dispose the feed bags, fertilizer bags, used bottles and containers

Record keeping

- ❖ Maintain records of all farm activities, stocking details, inputs, sampling details, harvest details, sale and consumption.
- ❖ Use record books provided under the project for keeping records.
- ❖ Ensure the records are updated periodically with the support of Project staff or Department officials during their visit.

Paddy-fish farming calendar

Follow the farming calendar presented below:

Months	Activities	
	Paddy farming	Fish farming
January	Sowing, Nursery raising and Transplanting	Civil works
February		Pond Preparation
March		Fish Seed Stocking
April		
May	Harvesting	
June		
July		Partial Harvesting
August		
September		
October		Fish Harvesting
November		
December		

Chapter 7



Quality carp seed production



7 Quality carp seed production



Introduction:

The aquaculture system in Assam is mainly categorized by polyculture of Indian Major Carps with Catla and Rohu as major species along with exotic carps & other minor carps. The availability of quality fish seed in adequate quantities throughout the year is the prime requisite for aquaculture development to increase production, vertical expansion and growth of the industry. Seed supply is a major problem faced by fish farmers of Assam in terms of quality, quantity and timely availability. The state has got around 500 carp hatcheries in both Government and private sector that produced around 95190 lakhs fish seed (fry) during the year 2019 – 20. However, due to lack of adequate nursery rearing ponds for regional self-sufficiency, breeding practices involving underweight and under size immature fishes, irregular brood stock replacement, improper maintenance of brood fish and mixed species breeding practices as well as lack of awareness on scientific management of fish seed rearing ponds, poor fish seed transportation and poor marketing facilities act as major problems for quality fish seed production. The concept of brood nourishment, brood stock management and use of improved strains are at the initial stages of adoption by the hatcheries. With the growing requirement for quality carp seed, there is increasing need for development of guideline or Standard Operating Procedure for Indian Major carps hatcheries in Assam context. Assam being the feeder state for fish seeds for the entire NE region, to cater the need of the growing fish farming sector of the region, the fish seed production sector of the state of Assam should be improved with technology for quality seed production in right time.

Pond preparation

- ❖ The brood stock ponds need to be prepared as per the SOP and brood stock managed with care for 6 - 8 months prior to spawning.
- ❖ Pond should be completely de watered (drained or pumped out), sun dried for a couple of days and then rake the soil for escaping of accumulated poisonous gas.
- ❖ In case it is difficult to completely drain the pond, at least $\frac{3}{4}$ should be dewatered and then manhua oil cake @ 250kg/ha can be applied. This initially acts as poison to control weed and predatory fishes and later as manure.
- ❖ The lime has to be applied @ 250- 500 kg/ha depending on the soil pH for disinfection of pond bottom and to act as a buffering agent to correct the pH.
- ❖ The pond then be filled with water up to 2 feet followed by Manuring with cattle dung @ 10000 kg/ha/year (50% initially and rest in equal installments).
- ❖ If open canal water is used, place a screen to the inlet to avoid entry of SIS or other weed/predatory fishes
- ❖ The ponds should be filled to a depth of five to seven feet (1.5-2.0 m).
- ❖ The brood stock can be released 7 days after application of manure @ 1000 kg/ha. (Two weeks in case if Manhua oil cake is used).

Brood fish selection and stocking

- ❖ Virgin young fish or fish at first maturity of minimum 1.5 years age for Rohu & Mrigal and 2.5 years age for Catla. Ideal size of fish at stocking should be Rohu 1-1.5 kg, Mrigal 0.8 – 1.2 kg and Catla 1.5 – 2.0 kg.
- ❖ Record the source of the brood stock (farm reared or from grow out tanks- if out sourced, origin of the seed may be recorded). Incorporate 30% from the early bred stock from outside by outsourcing into the brood stock). Avoid fish of unknown source, history and genetic makeup.
- ❖ Record weight and age of all the fish at stocking.
- ❖ Stocking density @ 2000kg/ha or 800 kg/acre.
- ❖ Sex wise separate stocking is preferred.

Feed management

- ❖ The brood fish should be raised in brood fish ponds with special care.
- ❖ The brood fish should be fed with brood feed (32/4) @ 2% of body weight for initial 3 months (up to November)
- ❖ For the months of Dec and Feb (winter months), feeding with brood feed (28/4) @ 0.5% of BW
- ❖ For rest of the period till spawning fish to be fed with same feed (28/4) @ 1% of BW
- ❖ Add Vitamin E with Selenium (Available in poultry input shops) @ 5ml/kg feed daily (start from Jan end or February)
- ❖ Stop feeding once in a week (fed only for 6 days in a week).
- ❖ In winter feed only 4 – 5 days in a week depending on demand.
- ❖ In case of presence/occurrence of small indigenous self-populating species (SIS), or any other unwanted fish, they should be removed by repeated netting as they compete with feed (both zooplankton and also supplementary feed given)
- ❖ If any infestation of *Argulus sp*, use insecticides as cypermethrin 3 times in 7 days intervals at evening time @ 0.5 ml/ft/dec water. If any infestation of *Lernaea sp*, use Dipterex 3 times in 7 days intervals at evening time @ 1ppm in pond water or can also give Bath treatment with 25mg/l of formalin or 2-3% NaCl for 15 min or 100ppm KMnO₄ pond treatment with 5mg/liter of KMnO₄

Water Quality management

- ❖ Monthly application of lime @ 60kg/ha
- ❖ Raw salt application @ 125kg/month/ha
- ❖ Exchange the water @ 20% - 30%
- ❖ Pull chain at the pond bottom once in two weeks for removal of poisonous gas
- ❖ Use zeolite @ 50kg/ha/month to improve the bottom ecology.
- ❖ Check the fish once in a month for monitoring growth and health check up
- ❖ Reschedule feeding based on the biomass as indicated

Breeding and Hatchery management

Hatchery preparation (prior to spawning)

Water source:

- Preferably ground water (tube well or bore well) is preferred and the water should be stored in a reservoir or pond/sump tank before pumping to overhead tank- aging process to increase the dissolved oxygen content in the water. If earthen pond is used for storing water, pond should be sun-dried for about a week, and then filled with clean water. Care should be taken to prevent entry of any fish or other unwanted animal by providing screen at entry or inlet point.
- If water from river or other natural sources is used, it should be stored in a clean reservoir/storage tank as indicated above.
- Overhead tank, and all pipelines and fittings and equipment should be checked, repaired, cleaned and disinfected using lime or KMnO_4 before use.
- Spawning and hatching pools should be thoroughly washed with limewater and disinfected with KMnO_4 .

Induced spawning steps:

- Brood fish selection and conditioning:
- Fully mature Brood fish, ready to spawn should be selected by observing secondary sexual characters –
- Females: Pectoral fin soft; swollen belly round in shape and soft to touch with reddish vent.
- Males: Pectoral fin rough with tubercles and oozing of milt with slight pressure on abdomen
- Net the brooders in the morning hours and condition them for spawning
- Select fish of 2 years or more of equal weight (preferably <3kg) and equalize sex ratio by selecting equal numbers fish of almost same age and equal weight
- The brood fishes should preferably be conditioned for 4 -6 hours before hypophysation in a separate concrete pond to empty stomach and acclimatization to the new hatchery environment
- Brood stock should be collected and transferred to hatchery in wet condition (with water or in wet cloth) for conditioning during morning cool hours and should be handled with care to avoid stress injuries. Avoid using brood stock of unknown source
- If the brood stock pond is far off from the hatchery, use mild sedative (one drop of clove oil/liter of water) to avoid or minimize the stress during transportation.

Spawning protocols:

- Spawning of different species in the same spawning pool simultaneously / mixed species breeding should be avoided to prevent possible cross breeding or hybridization as intergenetic hybridization is common in IMC.
- Evening hours (just before sun set-5.30—6.00 pm) is the best time for administering inducing agent.
- Weight of each male and female injected should be recorded along with numbers
- Females may be administered inducing agent first followed by males.
- Commercially available inducing agents (Ovaprim, Ovatile, WOVA-FH) may be used.
- Spawning pool should be covered with net covering to avoid possible escape of brood fish by jumping.
- Water circulation and showering in spawning pool and hatching pool should be at appropriate speed and continuous and maintain the water temperature around 30° C.
- Number of breeders injected should be based on the capacity of the spawning pool and the capacity of the hatching pools.
- Provision for power back for continuous pumping of water is essential.

Removal of spawned fish from the breeding pool and collection and transportation of eggs to hatching pools:

- The spawning takes place in the early morning hours and water hardening of eggs completes in 3-4 hours post fertilization.
- The eggs will be ready for shifting to hatching pools on the 2nd day morning.
- Brood fish should be removed from the breeding pool in the morning hours and disinfect them by dipping in KMnO₄ solution.
- Record the number and weight of females fully released the eggs-- calculate fecundity and fertilization success.
- Record the number and weight of females not responded and shift them to brood stock pond for preparing them for next spawning.
- Spawned fish or spent, both male and female may be shifted for a separate pond to nurse them for later use.

Incubation of eggs or hatching practices:

- The water-hardened eggs should be transferred into the hatching pools 6-7 hours after fertilization.
- Chinese type of hatcheries are preferred for commercial scale egg incubation.
- Maintain water temperature around 30-32° C through showers and bringing down water temperature by using ice blocks to overhead tank, if temperature is very high.

- Dissolved Oxygen concentration in hatchery water should be maintained at 5-7 ppm.
- Alkalinity of water should be maintained below 130 ppm.
- Remove the dead eggs/hatchlings periodically from the hatcheries to avoid infection to other eggs/ hatchlings.
- Water circulation rate in the hatching polls should be maintained at a proper rate throughout hatching period for oxygen supply and better Survival.
- Normally yolk resorption completes between 72-96 hrs. after hatching and spawn will be ready by 4th day evening or 5th day morning which has to be shifted for nursing in nursery ponds.
- Early Morning time (before 9 am) is the best time for collection of spawn for transportation and shifting to nursery rearing.
- Measure the quantity of spawn using measuring cups of known volume (measure randomly 3-4 cups, count and then compute to get total spawn obtained.
- Ideally, spawn should be conditioned in a small tank with water flow for one or two hours before transportation or shifting to nursery ponds where dead spawn can be removed and counting also can be done.
- Spawn packing and transportation.
- 50,000 spawn per polythene bag with 4 litres of water and 4 litres volume oxygen.
- Spawn should be ideally transported in cool hours –morning or evening hours

Nursing spawn for fry production (Nursery Pond management):

- ❖ Earthen or cement ponds of 0.1 ha with a Water depth of 1.00-1.20 m or 3-4 ft are ideal for spawn nursing.
- ❖ Ponds should be completely drained and sun dried (if drainable), rake the pond bottom for release of accumulated poisonous gases and oxidation of organic matter, and kill harmful bacteria.
- ❖ Removal of aquatic weeds and predators if any.
- ❖ Liming-@ 200 – 500 kg/ha depending on the soil pH. Using agriculture lime Optimal water pH is 6.5-8.5
- ❖ Cow dung and poultry manure 7-10 days ahead of stocking at the rate of 5 – 6 tons/ha and 2 – 3 tons/ha respectively.
- ❖ Spawn stocking density @ 4 million/ha
- ❖ Feeding 6 kg/million/day for the first 5 days. For the remaining period, 12 kg/million/day.
- ❖ The two feed ration/day is necessary to get greater survival rate and enhanced growth rate. For first few days give powdered feed or in liquid form by spreading across the pond.

- ❖ Number of cycles per season: 5 (making allowance of 5 days for preparation)
- ❖ Spawn nursing duration- 15-20days
- ❖ Spawn should be stocked in nursing tank either early morning or late evening hours.
- ❖ Nursing of single species is better as segregation of species at fry level will be extremely difficult

Chapter 8



Beel fisheries development



8 Beel fisheries development

Introduction:

Assam has the maximum water spread area under floodplain wetlands, mainly associated with the rivers Brahmaputra and Barak. Locally known as beels, they are mostly oxbow lakes, back swamps, dead rivers or tectonic depressions. The floodplain wetlands are productive ecosystems with an estimated fish production potential of 1500 kg/ha/yr. However, the current fish production from the beel is 335 kg/ha/yr. This is far less than their potential because of poor recruitment from the catchment areas, the rivers, canals and wetlands.

If managed along scientific lines, fish production in beels can be increased significantly. Therefore, in order to achieve the major objectives of increasing productivity, diversity in fish production and to benefit the beel user communities in a sustainable way it is important to follow the interventions on Beel Fisheries Management mentioned below:

Technical Interventions for Beel Fisheries Management

Weed Infestation:

- ❖ For beels weed infestation may be allowed up to 25% the total surface area.
- ❖ For maintaining productive status of the beels, the presence of less weeds is preferable.
- ❖ It is suggested to remove all the aquatic weeds except in few selected sites within the beel for habitat improvement by keeping limited volume of floating aquatic weeds or small proportion of weeds in the peripheral areas by erecting *banas* or by using ropes.

Enhanced Stocking:

Species of fishes to be stocked:

- ❖ In order to get good fish production, it is suggested to stock fingerlings of catla, rohu, mrigal, grass carp, common carp and minor carps.
- ❖ Small indigenous fishes like mola, puntius spp, etc are also recommended for stocking.

Stocking density:

- ❖ Recommended stocking density is 4000 Indian major carp fingerlings per ha.
- ❖ In addition stocking of minor carps and the small indigenous fishes recommended.

Stocking of mola and other Small Indigenous fishes:

- ❖ Stocking of mola and other small indigenous fishes in beels will help to get additional production of mola and other small indigenous species of fishes for local consumption by the beel user communities.
- ❖ The stocking density of mola and SIS in beel should be at the rate of 5 kg per ha.

Stocking Size of fish and Stocking time:

- ❖ In case of Indian major carps, it is recommended to stock large size fingerlings of >10 cm.
- ❖ Recommended seeds should preferably be stocked during the period from April to June to get better results.
- ❖ In situ rearing of fingerlings using nurseries, pens or cages is recommended to get supply of good quality and large size fingerlings.
- ❖ Stocking of large size fingerlings produced from their own rearing tanks and supplementary feeding would lead to increase production through harvest of large size fishes.

In Situ production of fingerlings

- ❖ For *in situ* production of fingerlings nursery ponds have to be constructed of by the side of the beel.
- ❖ Setting up of pens and cages for fingerling nursing within the beel areas.
- ❖ Cage culture in deeper beels and pen culture in shallower beels or marginal areas of the beels can be promoted; especially for producing good quality fingerlings of desirable sizes of different species of carps, which would be useful for stocking in the beels.
- ❖ In addition to large size fingerling, production in enclosures and hapas can be promoted for certain species like common carp.

Supply of large Size Fingerlings from Local Nurseries

- ❖ Where there is no scope to establish nurseries, pens or cages for large size fingerling production in or around the beel area, it is better to get the supply of large size fingerlings produced from nurseries established in the villages surrounding the beels.
- ❖ The promotion of such nurseries will be useful to support members of the BDMC (Beel Development Management committee) to get additional benefits from nursery operations.

Facilitate Self-recruitment of the native fishes

- ❖ Self-recruitment of the native species of fish in the beel should be facilitated by improving habitat, by providing sanctuaries within the Beel in the deepest area and through better connectivity to river and wetland system.

- ❖ Stocking of native species of fish (e.g. mola and other small indigenous fishes) would be effective to increase the production from beel.
- ❖ It is useful to take conservation measures to preserve the native species of fishes in beels by creating habitats and/or protected areas within the beels.

Creation of Sanctuary within the Beel

- ❖ It is useful to keep 5-10% of the area of the beel especially in the deeper areas as fishing-free zone for conservation of small indigenous species and other self-recruiting native species of fish.
- ❖ Of the sites developed as habitats for the fish in the deeper areas, 2-3 sites can be selected as sanctuaries by establishing *Kattals* using branches of trees, bamboos.

Harvesting of Fish

- ❖ Harvest large size fishes (Indian major carps) when they grow to marketable size of at least 1 Kg or above based on the market demand.
- ❖ Sometimes, due to late stocking, majority of the fishes will not be able to reach up to the marketable size, so in that case partial harvest of marketable size fishes could be useful.
- ❖ The partial harvesting of the fish will allow the rest of the fish to grow to marketable size within short period.
- ❖ The Beel Development Management Committee could use the strategy of multiple stocking if they have access to large size fingerlings available from their own nurseries, pens or cages.
- ❖ Harvesting of Mola and other Small Indigenous Fishes should be carried out after 3 months of stocking and may be continued for at least 5-6 months per year to get higher productivity and income.
- ❖ Transportation and sale of fish as fresh or live fish in the market would help to get premium price.

Monitoring and Record Keeping

- ❖ Record should be maintained regarding input use, expenditure, production, income, sharing of benefits and others.
- ❖ The spawn stocked, fry production, fingerlings production, number of fingerlings stocked in the beel, additional fingerlings available or sold need to be recorded.
- ❖ For production it is important to keep record of the details regarding fish production, species harvested, size of the fish harvested, price of fish sold by species, period of sale, quantity sold in the market, quantity of the harvested fishes consumed locally within the beel user communities, etc.

About WorldFish

WorldFish is a nonprofit research and innovation institution that creates, advances and translates scientific research on aquatic food systems into scalable solutions with transformational impact on human well-being and the environment. Our research data, evidence and insights shape better practices, policies and investment decisions for sustainable development in low- and middle-income countries.

We have a global presence across 20 countries in Asia, Africa and the Pacific with 460 staff of 30 nationalities deployed where the greatest sustainable development challenges can be addressed through holistic aquatic food systems solutions.

Our research and innovation work spans climate change, food security and nutrition, sustainable fisheries and aquaculture, the blue economy and ocean governance, One Health, genetics and AgriTech, and it integrates evidence and perspectives on gender, youth and social inclusion. Our approach empowers people for change over the long term: research excellence and engagement with national and international partners are at the heart of our efforts to set new agendas, build capacities and support better decision-making on the critical issues of our times.

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For more information, please visit www.worldfishcenter.org